PROJECT SUTRA — HARIKA'S MASTER PITCH SCRIPT

SPEAKER: HARIKA • OPERATIONS LEAD • PHASE 1: HOOK, PROBLEM & FULL SOLUTION

SLIDES 01 TO 03 (COMPLETE LEAD)
TOTAL TIME: ~1 MIN 50 SEC

💡 HARIKA'S STRATEGY: You deliver the entire opening arc — from the real disaster hook, through the 4 failure breakdowns, to the 4 engineering solutions and performance numbers. You establish complete authority, then execute the power handoff to Nikhil for technical deep dives and Q&A defense.

SLIDE 01 Opening Hook & Project SUTRA Introduction

00:00 – 00:30 (30s)

[Stand center stage • Take a 1-second breath • Deliver opening hook with serious, grounded tone]

🔥 **HOOK:** "When over 300 lives were buried under mud in the **July 2024 Wayanad landslides**, and catastrophic **Himalayan glacier bursts cut off entire river valleys across Kedarnath and Nepal**, rescue teams had only 72 hours to save lives. But the moment standard drones entered thick forests and mountain gorges, they lost signal and crashed."

"Respected jury, we are **Team OFFGRID**. I am Harika, and with me is Nikhil, our Tech Lead. We built **Project SUTRA** — an autonomous drone swarm engineered to find survivors even when **GPS fails and normal communications collapse**."

- PILLAR 1 **Self-Flying Swarms:** Drones dodge trees and coordinate together at 50 times a second without ground towers. [50Hz GNC]
- PILLAR 2 **Smart AI Video:** Live video stays connected through deep mountain valleys without blacking out. [-5dB JSCC]
- PILLAR 3 **Thermal & 3D AI Vision:** Spots survivors through smoke and finds their exact coordinates on steep slopes. [3D DEM • 3.59cm]

SLIDE 02 The Problem: 4 Real Disaster Breakdowns

00:30 – 01:05 (35s)

[Click to Slide 2 • Switch tone to urgent, factual disaster audit]

⚠️ **THE AUDIT: "When our team audited actual NDRF and Army rescue reports from recent disaster zones, we found 4 major breakdowns:"**

- BREAKDOWN 1 **Trees Block GPS:** In Wayanad, thick tree cover blocked satellite signals, causing **70% of drones to crash into branches**.
- BREAKDOWN 2 **Mountains Block Radio:** In mountain gorges, steep ridges blocked radio signals, dropping video feeds to total black.
- BREAKDOWN 3 **Slope Errors:** On steep 45-degree hillsides, standard drone cameras assumed flat ground, **missing survivor locations by 15 to 30 meters**.
- BREAKDOWN 4 **Pilot Overload:** Deploying 5 separate drones required **15 to 25 crew members** and 2 hours of setup, costing ₹12.5 Lakhs per flight.

SLIDE 03 The Solution: 4 Hardened Moats & Performance Delta

01:05 – 01:50 (45s)

[Click to Slide 3 • Deliver solutions with confident clarity • Point to performance table]

🛡️ **THE SOLUTION:** "SUTRA solves each breakdown through 4 hardened engineering moats:"

- MOAT A **Self-Flying Swarms:** Drones calculate collision-free flight paths onboard at 50 times a second — zero ground towers, zero GPS reliance.
- MOAT B **Smart AI Video Link:** When signals drop behind mountain ridges, our neural link keeps the video running down to -5dB without blacking out.
- MOAT C **Thermal & 3D AI Vision:** Combines body heat sensors with 3D elevation raycasting on Jetson Orin (<15ms), **completely removing the 30-meter slope error**.
- MOAT D **1-Person Command Tablet:** An offline 3D tactical map allows **just 1 operator to fly the entire swarm**, cutting mission costs to **₹95,000**.

Metric	Traditional SAR	Single Drone	SUTRA SWARM
Area Coverage / mi²	2-3 hours	45-60 min	10-18 min (4x Faster)
Detection Rate	65-70%	55-65%	78-85% First Pass
Personnel Required	15-25 people	2-3 pilots	1-2 Operators
Sortie Cost	₹12.5 Lakhs	₹5.0 Lakhs	₹95,000

"The bottom line: SUTRA searches a full square mile in just 10 minutes — 4 times faster than human rescue crews, saving lives when every second counts."

🤝 POWER HANDOFF TO NIKHIL:
"Now, our Tech Lead, Nikhil, will walk you through the deep engineering behind each subsystem and our real-world deployment."